

Tulga-Erdene Sodjargal

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Education

KAIST (Korea Advanced Institute of Science and Technology)

B.S. IN BIO AND BRAIN ENGINEERING, DOUBLE MAJOR IN CHEMISTRY

Daejeon, South Korea

Aug. 2021 – Feb. 2027 (expected)

- Cumulative GPA: 92.3/100 (as of Jul. 2026).
- Machine Learning & Statistics:** Statistical ML, Statistical Methods with Computers, Probability and Statistics, Biomedical Statistics and ML, Big Data and Machine Learning in Biotechnology.
- Bioinformatics & Computational Chemistry:** ML for Molecules and Materials (graduate), AI Chemistry, Computational Chemistry, Physical Chemistry I–II (quantum chemistry and statistical thermodynamics), Bioinformatics, Bio-Information Processing, Bio-Data Engineering, Bio-Data Structures.

École Polytechnique Fédérale de Lausanne (EPFL)

VISITING & PROJECT STUDENT

Lausanne, Switzerland

Sep. 2024 – Jun. 2025

- Exchange program with a part-time research project (Sep. 2024 – Feb. 2025), followed by a full-time research project in the Laboratory of Computational Science and Modeling (Prof. Michele Ceriotti; Feb. – Jun. 2025).

Skills

Programming	Python, R, MATLAB, Bash, SQL
ML & Deep Learning	PyTorch, TensorFlow, scikit-learn, Optuna
Molecular & Scientific Computing	ASE, LAMMPS, NAMD, AutoDock Vina, Gaussian, MDAAnalysis, RDKit
Data & Software Engineering	NumPy, pandas, PostgreSQL, Git, CI/CD, pytest
Wet-Lab Techniques	Air-sensitive reactions, flash chromatography, $^1\text{H}/^{13}\text{C}$ NMR (1D, 2D)
Languages	English (TOEFL iBT 116/120), Russian (proficient), Mongolian (native), Korean (working)

Publications & Presentations

JOURNAL & PEER-REVIEWED

Learning Long-Range Representations with Equivariant Messages

E RUMIANTSEV, MF LANGER, **T-E SODJARGAL**, M CERIOTTI, P LOCHE

Transactions on Machine Learning Research (TMLR)

Journal

2025

scicode-widgets: Bringing Computational Experiments to the Classroom with Jupyter Widgets

A GOSCINSKI, TJ BAIRD, D DU, J PRADO, D SUMAN, **T-E SODJARGAL**, S BONELLA, G PIZZI, M CERIOTTI

arXiv:2507.05734 (under review)

Preprint

2025

CONFERENCE PRESENTATIONS

Integrating Long-Range Interactions into Machine Learning Interatomic Potentials

T-E SODJARGAL, E RUMIANTSEV, P LOCHE, M CERIOTTI

German Physical Society (DPG) Spring Meeting, session MM 9.62 (first author)

Poster

2025

Experience

Laboratory of Computational Science and Modeling, EPFL (Prof. Michele Ceriotti)

RESEARCH INTERN (ATOMISTIC ML)

Lausanne, Switzerland

Sep. 2024 – Sep. 2025

- Developed machine-learning interatomic potentials that incorporate long-range interactions, improving accuracy for molecular property prediction and molecular dynamics.
- Co-authored a journal paper (*TMLR*, 2025), presented first-author results as a poster at the German Physical Society (DPG) Spring Meeting 2025.
- Contributed to open-source atomistic-ML libraries and interactive Jupyter-based teaching materials (preprint under review in *Journal of Open Source Education*), including PyTorch-level optimizations and CI/CD.
- Tools:** PyTorch, ASE, LAMMPS, ipywidgets, Git, Bash

University of Illinois Urbana-Champaign (Prof. Taras V. Pogorelov)

RESEARCH INTERN (MOLECULAR DYNAMICS)

Remote
Feb. 2024 – Present

- Analyze all-atom molecular-dynamics trajectories of cellular membranes of differing lipid composition to support multi-team drug-design projects.
- Refactored the analysis pipeline, cutting runtime by $> 30\times$ and enabling structural insights not previously tractable.
- Co-authoring a manuscript currently in preparation.
- Tools:** MDAnalysis, scikit-learn, NumPy

Wellman Center for Photomedicine, Harvard Medical School (Prof. Mei X. Wu)

RESEARCH INTERN (COMPUTATIONAL BIOCHEMISTRY)

Boston, USA
Jun. 2025 – Aug. 2025

- Established the group's computational research capability as its sole dry-lab researcher within an otherwise wet-lab team.
- Independently designed and ran molecular-docking and simulation studies to rationalize and guide experimental results.
- Tools:** AutoDock Vina, NAMD, Gaussian, Bash

PuzzleAI

RESEARCH INTERN (LLM DATA ENGINEERING)

Daejeon, South Korea
Jan. 2026 – Feb. 2026

- Built a robust pipeline for LLM-driven single-turn dataset expansion, with automated Korean-to-English translation and I/O parsing for downstream training.
- Tools:** OpenAI API, Python

SpiderCore Inc.

RESEARCH INTERN [BACHELOR'S THESIS] (ML FOR MOLECULES)

Daejeon, South Korea
Jun. 2024 – Aug. 2024

- Built graph neural networks for gene-therapy design, contributing domain-specific chemical expertise to model design.
- Devised a chemically-inspired self-supervised pretraining task that lifted model performance to state-of-the-art levels.
- Tools:** TensorFlow, RDKit

KAIST

UNDERGRADUATE RESEARCHER

Daejeon, South Korea
Mar. 2022 – Nov. 2023

- Designed candidate therapeutic antibody variants using deep-learning protein-design tools (Prof. Byung-Ha Oh; Jun. – Nov. 2023).
- Developed novel chemical reactions via combinatorial screening (Prof. Yoonsu Park; Mar. 2022 – Jun. 2023).
- Tools:** RFdiffusion, ProteinMPNN, Bash

Selected Projects

Cost-Aware Demand Forecasting for Electronic Parts

3RD POSTECH-UNIST-KAIST DATA SCIENCE COMPETITION

- Developed time-series forecasting models with cost-aware objective optimization.
- Silver Award, 5th of 20+ teams from leading Korean universities.
- Tools:** Nixtla, pandas

Predicting Hydrogen-Bond Basicity (pK_{BHX}) on a Small Dataset

FINAL PROJECT, *AI Chemistry*

- Built a graph convolutional neural network to predict hydrogen-bond basicity from molecular structure on a low-data regime (~350 molecules), using regularization and hyperparameter tuning to counter overfitting.
- Ranked 1st among undergraduates (4th of 13 overall) in the course-wide challenge.
- Tools:** PyTorch, RDKit

Housing Price Prediction

FINAL PROJECT, *Statistical ML*

- Built regression models with feature engineering, preprocessing, and hyperparameter tuning; ranked 7th of 60 in the class competition.
- Tools:** scikit-learn, pandas, Optuna

Analyzing Workplace Discrimination in Korea

FINAL PROJECT, *Statistical Methods with Computing*

- Uncovered national trends in workplace discrimination through EDA, hypothesis testing, and clustering.
- Tools:** scikit-learn, pandas

SNP Analysis of the COVID-19 Delta Variant Surge

FINAL PROJECT, *Bio-Data Structures*

- Implemented a heuristic global sequence-alignment algorithm from scratch in pure Python to detect SNPs associated with the Delta variant surge in England.
- Interpreted the biological roles of identified SNPs through targeted literature review.

Biomedical Information System for Healthcare

FINAL PROJECT, *Bio-Information Processing*

- Designed a relational database integrating genomics and pharmacokinetics data, with a command-line interface simulating healthcare-provider use cases.
- Tools:** PostgreSQL, psycopg2